

CATERPILLAR® MACHINE DIAGNOSTICS

INDUSTRIAL INSPECTION & SYSTEM CONFIGURATION AUDIT

Machine ID:	CAT-BDZ-D6	Date of Inspection:	2026-07-20
Machine Name:	CAT Bulldozer	Lead Engineer:	Sarah Engineer
Manufacturer:	Caterpillar Inc.	Model / Category:	D6-LGP / CAT Bulldozer
Diagnostic Status:	Fault (Score: 70%)	Engine Version:	v2.4-deterministic

System Hardware & Firmware Configuration Audit

Below is a side-by-side comparison of the active telemetry and local software modules against original factory reference parameters.

Parameter	Factory Reference	Current Machine	Audit Status
Firmware Version	v6.1.0-build22	v5.9.0-build11	MISMATCH (CRITICAL)
PLC Version	v5.0.1	v5.0.1	MATCHED
CPU Architecture	AMD G-Series GX-412	AMD G-Series GX-412	MATCHED
RAM (Memory)	4GB DDR3	4GB DDR3	MATCHED
Storage	32GB mSATA	32GB mSATA	MATCHED
Sensor Count	16	14	MISMATCH (WARNING)
Communication Ports	USB, COM1, Ethernet, Modbus-RTU	USB, COM1, Ethernet, Modbus-RTU	MATCHED
Installed Modules	Analog Input, CAN Bus controller, GPS receiver	Analog Input, CAN Bus controller, GPS receiver	MATCHED
Operating Temperature	Under 70 C	68.2 C	MATCHED
Power Supply Status	Stable	Stable	MATCHED

Detected Mismatches & Anomalies

1. Firmware Version Mismatch [CRITICAL]

Firmware version mismatch. Expected 'v6.1.0-build22', found 'v5.9.0-build11'.

2. Sensor Count Mismatch [WARNING]

Active sensor count mismatch. Expected '16' nodes, detected '14' nodes.

AI-Powered Diagnostics Analysis

Machine Safety & Health Assessment

DEGRADED / ACTION REQUIRED (Health: 70%): Mismatches or anomalous sensor readings detected. Operational risks present.

Root Cause Analysis

The diagnostic engine flagged the following configuration mismatches: - Firmware Version: Firmware version mismatch. Expected 'v6.1.0-build22', found 'v5.9.0-build11'. - Sensor Count: Active sensor count mismatch. Expected '16' nodes, detected '14' nodes. Typical root causes include: field component replacement with unauthorized modules, out-of-date PLC programming flashes, firmware regression during maintenance cycles, or thermal cooling degradation.

Suggested Maintenance Worksteps

1. Connect Service Tool (CAT ET or Siemens TIA Portal) and flash target firmware and PLC image profiles matching reference configurations.

Crucial Safety Notes

- Verify power source disconnect before opening PLC enclosures.
- Ensure constant power backup is connected to the PLC logic controllers during the programming and flashing sequence.

Verification & Troubleshooting Steps

1. Run firmware CRC checksum validation command. Check serial console output for boot loader warnings.

PREPARED BY:

Automated Diagnostic Engine
CAT Diagnostic Systems

REVIEWED & SIGNED BY:

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